

东亚时区RISC-V双周会

2026年02月05日·第 120 次

<https://github.com/cnrv/RISCV-East-Asia-Biweekly-Sync>

Host: 蒙卓

Organizer: PLCT Lab plct-oss@iscas.ac.cn

会议议程(15:00 - 16:00)

- 自我介绍、等待参会者接入、非技术话题八卦(没有的话就直接跳过)
- RVI 的更新和八卦(基本上跟东亚双周会群内消息同步)
- 东亚地区小伙伴的项目更新
- 自由讨论

RISC-V International 同步、全球开源社区八卦(陈逸轩)

[sig-academia]发布新课程 [Port Software to RISC-V](#)

[sig-perf-analysis][指令周期的堆栈](#)

[sig-soft-cpu]提议改为月度会议, 时间变成一小时。[总结2025, 展望2026](#)

[tech-crypto-ext]开立内存加密上下文任务组

[privileged-software]发布了 IOMMU spec 的 [clarification](#)

RISC-V 中文社区的同步与八卦(张宇溪)

- [一图看懂 K3 发布会](#)
- [MilkV Jupiter 2: 首个RVA23 RISC-V SBC](#)
- [率先完成适配！openKylin全面支持全球首款RVA23芯片K3](#)
- [北京人形天工平台核心运控系统率先完成国产RISC-V芯片验证](#)

- [算力融合新突破:openKylin基于超睿DP1000芯片与 KVM, 将OpenStack全栈移植RISC-V平台](#)
- [国内首款RISC-V AI算力芯片大规模量产！类TPU架构迎爆发](#)

- [赋能全球RISC-V形式化验证生态建设: 阎明铸当选Sail RISC-V Model项目Collaborator](#)
- [中国科学院软件研究所携手超睿科技和芯瞳实现全国产高性能 OpenHarmony 6.0 方案](#)

RISC-V 德语社区的同步与八卦

- [Security researchers at Germany's CISA Helmholtz Center for Information Security have found current RISC-V CPU implementations coming up short for their actual security.](#)

德国CISA亥姆霍兹信息安全中心的研究人员在FOSDEM 2026上报告指出RISC-V商用CPU实现存在显著安全短板，并未能如预期那样规避x86等架构曾面临的安全问题。

- 实际存在漏洞:即使是“简单”的顺序执行设计也存在架构缺陷和强大的侧信道漏洞，如 Cache+Time、CycleDrift
- 发现严重漏洞:通过其开源工具 RISCover，发现了人工分析遗漏的高危漏洞，例如平头哥玄铁C910中的“GhostWrite”漏洞，允许非特权代码直接写入物理内存，绕过内存隔离。
- 实现问题突出:问题根源主要在于厂商的实现选择，如启用不安全的默认配置、引入未经验证的私有扩展、缺少对推测执行的限制等。
- 补丁滞后:针对Spectre等瞬态执行攻击的Linux内核缓解补丁在RISC-V生态中落地也相对滞后。
- 紧迫的转折点:RISC-V正处于快速普及的关键期，当前厂商做出的不安全设计决策可能被固化到未来数十亿不可修补的芯片中。

- [TUM Unveils EU's 1st 7nm AI Chip with Local Processing and RISC-V Architecture](#)

慕尼黑工业大学(TUM)宣布成功研制出欧盟首款基于先进7纳米工艺的AI芯片，基于开源的RISC-V架构设计，属于神经形态芯片。其设计符合台积电的产业标准，计划自2028年起由欧洲半导体制造公司(ESMC)在德国德累斯顿进行生产。

- [Europe Microcontroller Market Report](#)

Industry Analysis From 2026 to 2034, 关于欧洲微控制器 (Microcontroller, MCU) 市场的核心内容总结, 报告预测期为2026年至2034年。主要增长驱动力为汽车行业变革, 工业自动化与能效要求, 边缘AI与物联网扩展

项目	数据
2025年市场规模	28.1亿美元
2026年预估规模	29.5亿美元
2034年预测规模	43.1亿美元
预测期年均复合增长率	4.86%
最大细分市场 (按类型)	32位微控制器 (2024年占58.3%)
最大细分市场 (按应用)	工业领域 (2024年占31.2%)
主导国家市场	德国 (2024年占28.2%)

RISC-V 德语社区的同步与八卦

- [The OpenHW Foundation unveils the first industry-ready RISC-V ecosystem to advance European digital sovereignty](#)

由OpenHW基金会牵头的“统一RISC-V IP访问平台(UAP)”正式发布, 这是欧洲首个为推进数字主权而构建的、行业就绪的RISC-V生态系统。UAP是欧盟投资数亿欧元的“TRISTAN”旗舰项目的直接产出。该项目于2023年启动, 汇聚了46个合作伙伴(包括恩智浦、Thales、西门子等巨头及顶尖研究机构), 目标是将RISC-V从研究环境推向产业化应用。The TRISTAN project is supported by Chips Joint Undertaking (CHIPS-JU) and its members Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, **Germany**, Denmark, Estonia, Greece, Spain, Finland, France, Hungary, Ireland, Iceland, Italy, Lithuania, Luxembourg, Latvia, Malta, Netherlands, Norway, Poland, Portugal, Romania, Sweden, Slovenia, Slovakia, Turkey.

- [A reliable supervisor system utilizing an FD-SOI FPGA and MRAM for COTS-based payload data processing](#)

Fraunhofer Institute for High-Speed-Dynamics, Ernst-Mach-Institut, EMI, Freiburg, Germany

提出了一款用于小型卫星成像载荷的数据处理单元(DPU)的监控系统。处理器基于FPGA的软核RISC-V, 操作系统Zephyr RTOS(实时操作系统), 该系统已在2026年发射的微纳卫星上进行了实际部署, 是航天领域应用开放架构的典型案例。

Clang/LLVM 上游进展

- [RISC-V][MC] Add a RegisterClass definition for RVY (CHERI)
<https://github.com/llvm/llvm-project/commit/5eb3969c7a0b>
- [RISCV] Run VLOptimizer right after ISel
<https://github.com/llvm/llvm-project/commit/0d11f68a8aa2>
- [RISC-V][Mach-O] Add codegen support for Mach-O object format.
<https://github.com/llvm/llvm-project/commit/c6086dd5508>
- [RISCV] Remove experimental from Zalasr
<https://github.com/llvm/llvm-project/commit/a43b55edf42b>
- [RISCV][llvm-objdump] Support --symbolize-operands
<https://github.com/llvm/llvm-project/commit/5b4a5cf51f37>

GCC 进展

psABI for Hardware Control-Flow Integrity

https://docs.google.com/presentation/d/1IHj3yi2BJfxn_Sw1COSpczoGCBt6K77qbXtIPQIRAv0/edit?usp=sharing

Spacemit-X60 gcc cost model支持

<https://patchwork.sourcware.org/project/gcc/patch/DU0PR08MB9910AD914403747DD4FE7660B098A>

添加Svrsw60t59b扩展最小支持

<https://patchwork.sourcware.org/project/binutils/patch/20260203023726>

QEMU/Spike 进展(泽文)

[\[RFC PATCH v5 0/7\] riscv: add initial sdxext support - Chao Liu](#)

[\[PATCH v3 0/2\] QTest: Add RISC-V IOMMU bare-metal test using iommu-testdev - Chao Liu](#)

[\[PATCH v5 0/5\] Add support for K230 board - Chao Liu](#)

OCP(FP8/FP4):

[\[PATCH v3 13/19\] target/riscv: Expose Zvfofp8min property - Max Chou](#)

RISC-V RVSP:

[\[PATCH v5 0/4\] hw/riscv/virt: Add ACPI GED and powerdown support - liu.xuemei1](#)

[\[PATCH RFC 0/7\] RISC-V: Add support for ACPI RQSC table - Drew Fustini](#)

[Call for GSoC internship project ideas](#)

QEMU 训练营 2026 筹备阶段:

[Learning QEMU Camp Tutorial\(RISC-V + GPGPU\)](#)

[1st QEMU Training Camp 2026 Coordination Meeting a Success](#)

Sail/ACT 上游进展

#1507 Add the Zibi extension.

#1478 Add unratified Zvabd extension

#1520 Add LCOFI interrupt missed in Sscofpmf.

#1491 Add configuration option to handle reserved dynamic rounding modes.

#1462 Add configuration option for reserved behavior: odd number register for RV32Zdinx

#907 [ACT4] Update `run\_tests.py` script to avoid breaking terminal input
#906 [ACT4] Enable Spike tests in CI
#905 [ACT4] Update flags to work with latest version of Spike
#901 [ACT4] Spike config and trickbox
#899 [ACT4] Refactor configs
#896 [ACT4] Remove unused PRIV_TEST_EXCLUDE_COMPONENTS
#895 [ACT4] Remove `ref\_model\_type` from config files
#892 [ACT4] Move `RVMODEL\_BOOT` to end of test and jump to it
#891 [ACT4] Fix for c.nop bug in LA macro
#889 S Normative rule to coverpoint mapping
#888 [ACT4] Update UDB submodule to latest and point at new repo
#887 [ACT4] Update URLs to point to new riscv-unified-db repo
#886 [ACT4] Update extension parsing logic
#885 Cover reserved encodings of dynamic floating-point rounding modes in Ssstrict
#883 Vector normative rule mapping template
#882 fixed ifdef generation in vector_testgen
#881 Vector coverpoints
#880 InterruptsSm
#879 Adding rv64gcv reference config for vector
#876 Added Zalrsc, Zaamo coverpoints. Adjusted parameter generator to inc...
#875 [ACT4] Add option to exclude parts of priv testsuite name in extensions list
#874 RVA23 Atomics
#871 [AC4] Add time to CTP release title to ensure they stay ordered
#869 [ACT4] Initial Developer Docs
#866 [ACT4] Update UDB version
#864 Exception and Interrupt normative rule to coverpoint mappings
#863 [ACT4] Include M extension when V or Zvfh is present
#862 Rough draft CRD for RV120 and MC100
#859 (ACT4) Initial coverpoints for SvH
#431 Add tests for Smrnmi extension
#428 Add Zaamo subcomponent to A
#409 Add menvcfg support (Zicbom_Zicboz_Zicsr_Ssdtsso_Sstc)

V8 for RISC-V 更新(邱吉、陆亚涵)

1. 修复随机崩溃 (by wangruikang@iscas.ac.cn)
7540554: [riscv] Fix sp handling in MacroAssembler::LeaveFrame |
<https://chromium-review.googlesource.com/c/v8/v8/+7540554>
2. 模拟器中增加对sp相关地址访问check
7543083: [riscv] Add check stack pointer when access memory |
<https://chromium-review.googlesource.com/c/v8/v8/+7543083>
3. 实现select IR的指令选择
7515447: [riscv] Implement word64/word32 Select for Tst/CmpZero operations |
<https://chromium-review.googlesource.com/c/v8/v8/+7515447>
4. 修复RVV非法指令问题 (Zhijin Zeng zhijin.zeng@spacemit.com)
7502635: [riscv] Fix illegal instruction caused by unsupported SEW and LMUL configuration |
<https://chromium-review.googlesource.com/c/v8/v8/+7502635>

Port Upstream

5. 7543084: [riscv][baseline] Inline fast ToBoolean checks for Smis and static roots |
<https://chromium-review.googlesource.com/c/v8/v8/+7543084>
6. 7540545: [riscv][ic] Introduce CallNamedInterceptorSetter builtin |
<https://chromium-review.googlesource.com/c/v8/v8/+7540545>
7. 7538513: [riscv][wasmfx] Implement cont.bind. | <https://chromium-review.googlesource.com/c/v8/v8/+7538513>
8. 7529664: [riscv][cleanup] Use EnumSet<CpuFeature> in multiple places |
<https://chromium-review.googlesource.com/c/v8/v8/+7529664>

Go community work update (PLCT 蒙卓)

TL;DR Summary:

- RVV runtime optimization reviewing
- RV Zk asm support upstreaming, runtime/crypto library TBD
- runtime/secret upstreaming
- Fixing <https://github.com/golang/go/issues/76816> for RVA22U64 builder
- PIE internal link support

1. Authored/Co-authored Go-mainline CLs:

- 741860: cmd/link: add R_RISCV_TLS_IE support for internal link | <https://go-review.googlesource.com/c/go/+741860>
- 742200: cmd/link: enable PIE internal link for riscv64 | <https://go-review.googlesource.com/c/go/+742200>
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2. Reviewed Go-mainline CLs:

- 652717: doc, cmd/internal/obj/riscv: document the riscv64 assembler | <https://go-review.googlesource.com/c/go/+652717>
- 646736: internal/bytealg: vector implementation of equal for riscv64 | <https://go-review.googlesource.com/c/go/+646736>
- 646737: internal/bytealg: vector implementation of compare for riscv64 | <https://go-review.googlesource.com/c/go/+646737>
- 670875: riscv64: fix the path to the RISC-V extensions in spec.go | <https://go-review.googlesource.com/c/arch/+670875>
- cmd/compile: line number debug info regression in go1.25 around literal rewriting | <https://github.com/golang/go/issues/74576>
- 348389: cmd/compile: emit classify instructions for infinity tests on riscv64 | <https://go-review.googlesource.com/c/go/+348389>
- 670875: riscv64: fix the path to the RISC-V extensions in spec.go | <https://go-review.googlesource.com/c/arch/+670875>
- 690495: runtime: identify virtual memory layout for riscv64 | <https://go-review.googlesource.com/c/go/+690495>
- 703715: cmd/compile/internal/ssa: add codegen for Zicnd extension on riscv64 | <https://go-review.googlesource.com/c/go/+703715>
- 705996: cmd/compile/internal/ssa: add codegen for Zicnd extension on riscv64 | <https://go-review.googlesource.com/c/go/+705996>
- 734081: riscv64: fix bug in memOrder to string | <https://go-review.googlesource.com/c/arch/+734081>
- 671275: crypto/sha512: improve performance of riscv64 assembly when Zbb extension available | <https://go-review.googlesource.com/c/go/+671275>
- 717980: runtime, cmd/internal/obj/riscv: support spin lock on riscv64 | <https://go-review.googlesource.com/c/go/+717980>
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RuyiSDK (xijing)

- **RuyiSDK 包管理器**: RuyiSDK 0.45 已于今日发布, 对应的包管理器版本也为 0.45.0。
 - 由 pipx 安装的 ruyi 可以正常创建虚拟环境了。
 - 包管理器(本体部分)现已支持简体中文本地化。
- **RuyiSDK 软件源**: 完善了设备支持:
 - Milk-V Meles 的 RevyOS: 更新到 20251226。
 - Milk-V Pioneer 的 RevyOS: 更新到 20251226。
 - Sipeed Laptop 4A 的 RevyOS: 更新到 20251226。
 - Sipeed LicheePi 4A 的 RevyOS: 更新到 20251226。
 - Sipeed LicheePi Console 4A 的 RevyOS: 更新到 20251226。
 - Sipeed LicheeRV Nano 的 Buildroot SDK: 更新到 20251230。
- **RuyiSDK VSCode 插件**: 发布了 0.1.1 插件, 欢迎在VSCode插件市场搜索ruyisdk安装使用。
 - 解压包模板功能添加suggested folder(一键选择解压到当前目录或子目录)。
 - 重构虚拟环境功能。
 - 修复vscode marketplace的图片失效问题。
- **RuyiSDK Eclipse 插件**: 插件已经更新到 0.1.2 (测试中), 欢迎在Eclipse Marketplace中搜索ruyisdk或者riscv可搜索到插件。
 - 修复新闻模块第三方库的依赖问题。
 - 虚拟环境模块支持自动探测工作区中的项目。
 - 测试 Update Site 上架 Eclipse Marketplace 。Eclipse > Help > Eclipse Marketplace > Search

openEuler RISC-V (周嘉诚) (Offline)

Status / 20260205

- [openEuler 24.03 SP3:](#)
 - RVA23: Rebuilding & testing RC artifacts
 - RVA20: Rebuilding & testing RC artifacts
- Updates
- Kernel (RVCK 6.6)
 - Staging: fix for K3 system hang when bootup
 - Merged: revert a commit causing booting issues on SG2042
- GCC 12: Merged: backported fixes for AMO atomic ordering
- nodejs: Merged: backported fix for random crashes

Schedule of releases:

- [Early Q1](#) - openEuler [24.03 SP3 \(RVA23&20\)](#)
- [Late Q3](#) - openEuler [26.09 \(RVA23&20\)](#)
- [Late Q4](#) - openEuler [24.03 SP4 \(RVA23&20\)](#)

Features:

- 6.6-based [common kernel](#) for QEMU, Pioneer(SG2042), LPi4A(TH1520), BPi-F3(K1)
- UEFI-supported Hardware & QEMU images

Images:

- [UEFI ISO](#)
- [UEFI](#) qcow2 Image
- U-Boot Images for devboards

Fedora on RISC-V status update(20260205) 提交人不在线

- RPM packaging (<https://www.fedoravforce.org>)

- Koji Status: [F43, GA on October 28, 2025](#)

- [F43/rawhide: 17031 \[88.05%\] srpm](#)

- [RVA23\[ONGOING\]](#)

- [F42: 22471 \[92.32%\] srpm](#)

- main package version(F43):

- Toolchain:

- gcc-15.2.1-4

- glibc-2.42-4

- binutils-2.45.1-1

- libffi-3.5.2-1

- java-25-openjdk-25.0.0.0.36

- java-latest-openjdk-25.0.0.0.36

- perl-5.42.0-520

- python3.14-3.14.0-2

- llvm-21.1.6-1

- golang-1.25.2-1

- rust-1.91.1-1

- Desktop support Fedora 43:

- DONE:

- XFCE/Sugar/i3/Sway/GNOME/LXDE/KDE/LXQT

- ONGOING: Cinnamon/Budgie/Mate/Deepin

- Key Desktop App

- firefox-145.0.1-1

- libreoffice-25.8.3.2-3

- thunderbird-145.0-1

- chromium-137.0.7151.119-1.rv64

- Image and REPOs :

- <https://images.fedoravforce.com>

- Images:

- rsync://mirror.iscas.ac.cn/fedora-riscv/releases/43/Spins/

- REOP:

- rsync://mirror.iscas.ac.cn/fedora-riscv/releases/43/Everything

- ROS/ROS2 upgrading to F43

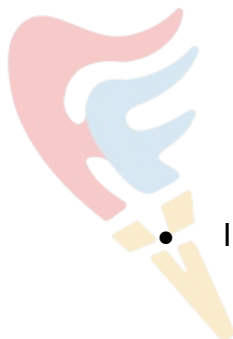
- [Sail](#) for rawhide[UPSTREAMING]

- function testing for F43:

- Podman, Image: [fedorariscv/base](#)

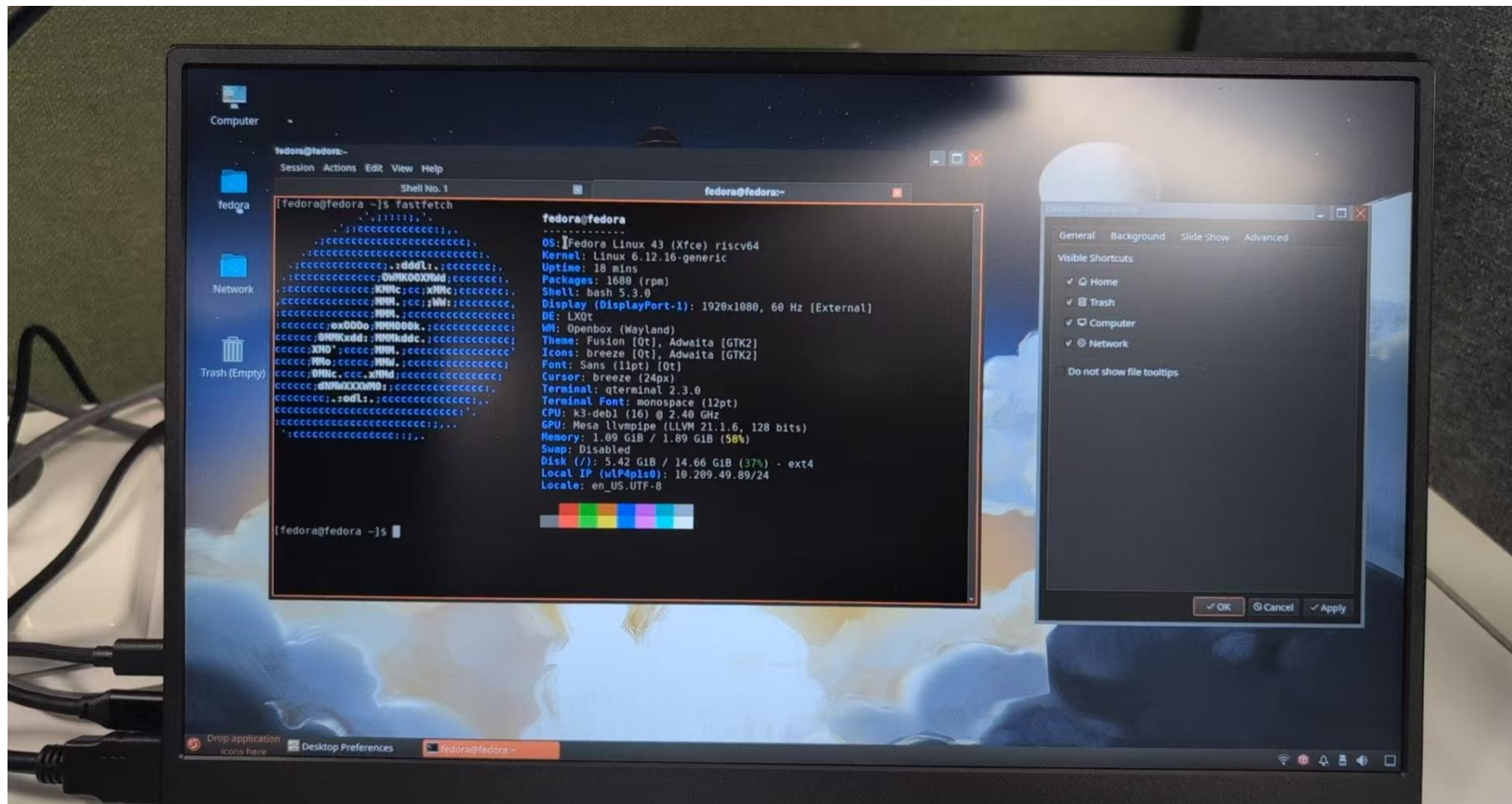
- Ceph[ONGOING] , DPDK, SPDK

- K8s[ONGOING]



Fedora-V Force

Fedora on RISC-V on SpacemiT K3 提交人不在线



固件（王翔）

- OpenSBI
 - 引入抽象的VIRQ
<https://lists.infradead.org/pipermail/opensbi/2026-February/009396.html>
- coreboot
 - coreboot引导riscv的UefiPayloadPkg的支持
<https://review.coreboot.org/c/coreboot/+91020>
<https://review.coreboot.org/c/coreboot/+91021>
<https://review.coreboot.org/c/coreboot/+91022>
- EDK2
 - 修正UefiPayloadPkg生成的Fit被dtc处理时报错的问题
<https://github.com/tianocore/edk2/pull/12062>
 - UniversalPayloadBuild.py 指定加载地址添加十六进制支持
<https://github.com/tianocore/edk2/pull/12061>
 - UefiPayloadPkg添加riscv支持, 现在可以进入uefi shell
<https://github.com/tianocore/edk2/pull/12061>

自由讨论 / AOB